

L07: Fundamental & Essential Matrix Direct Solution

8-Point-Algorithm: Direct Solutions for Computing Fundamental and Essential Matrix

- Compute F given ONLY corresponding points in images

- **Given:** N corresponding points

- $(x', y')_n, (x'', y'')_n$ with $n = 1, \dots, N$

- **wanted:** fundamental Matrix F

- Coplanarity constraint for each point:

- $\mathbf{x}_n'^T \mathbf{F} \mathbf{x}_n'' = 0 \quad n = 1, \dots, N$

- or

- $\begin{bmatrix} x'_n & y'_n & 1 \end{bmatrix} \underbrace{\begin{bmatrix} F_{11} & F_{12} & F_{13} \\ F_{21} & F_{22} & F_{23} \\ F_{31} & F_{32} & F_{33} \end{bmatrix}}_{\text{unknowns!}} \begin{bmatrix} x''_n \\ y''_n \\ 1 \end{bmatrix}$

- Linear Function in the unknowns F_{ij}

$$x''_n F_{11} x'_n + x''_n F_{21} y'_n + \dots = 0$$

$$\underbrace{x''_n x'_n}_{const.} F_{11} + \underbrace{x''_n y'_n}_{const.} F_{21} + \dots = 0$$

•

$$\begin{aligned} \mathbf{a}_n^\top &\longrightarrow [x_n'' x_n', x_n'' y_n', x_n'', y_n'' x_n', y_n'' y_n', y_n'', x_n', y_n', 1] \cdot \\ \mathbf{f} &\longrightarrow [F_{11}, F_{21}, F_{31}, F_{12}, F_{22}, F_{32}, F_{13}, F_{23}, F_{33}]^\top = 0 \\ n &= 1, \dots, N \end{aligned}$$



$$(\mathbf{x}_n'' \otimes \mathbf{x}_n')^\top \text{vec} \mathbf{F} = \underbrace{\mathbf{a}_n^\top}_{(\mathbf{x}_n'' \otimes \mathbf{x}_n')^\top \text{vec} \mathbf{F}} \underbrace{\mathbf{f}} = 0 \quad n = 1, \dots, N$$

(it holds in general: $\mathbf{x}^\top \mathbf{F} \mathbf{y} = (\mathbf{y} \otimes \mathbf{x})^\top \text{vec} \mathbf{F}$)

-
- Consider **all N points**
 - Directly obtaining a linear system

$$\underbrace{\mathbf{a}_n^\top}_{(\mathbf{x}_n'' \otimes \mathbf{x}_n')^\top \text{vec} \mathbf{F}} \underbrace{\mathbf{f}} = 0 \quad n = 1, \dots, N$$

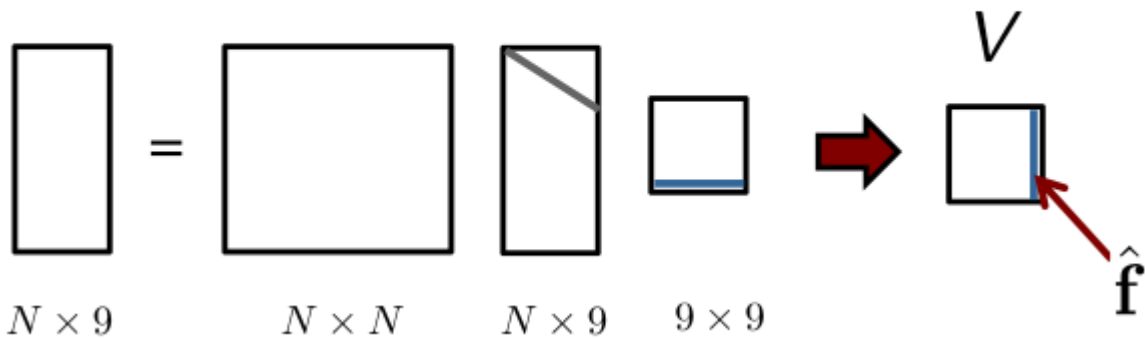


$$A = \begin{bmatrix} \mathbf{a}_1^\top \\ \vdots \\ \mathbf{a}_n^\top \\ \vdots \\ \mathbf{a}_N^\top \end{bmatrix} \quad \Rightarrow \quad A\mathbf{f} = \mathbf{0}$$

◦

- **Singular Value Decomposition**
- $\rightarrow$ Solve $A\mathbf{f} = \mathbf{0}$ for $\mathbf{f} = [F_{11}, F_{12}, F_{13}, F_{21}, F_{22}, F_{23}, F_{31}, F_{32}, F_{33}]$
- SVD: $\mathbf{f}$ is the right-singular vector corresponding to a singular value of A that is zero
- A has at most rank 8:
 - ≥ 8 **corresponding Points needed**
 - In reality: Noisy measurements
 - ≥ 8 points $\rightarrow$ A will become regular (should not!)
 - Singular vector $\hat{\mathbf{f}}$ of A corresponding to
 - **Smallest** singular value for solution $\hat{\mathbf{f}} \rightarrow \hat{\mathbf{F}}$

$$A = UDV^T$$



- Enforce $\text{rank}(F) = 2$
 - F should **approximate** computed matrix $\hat{F}$ as close as possible
 - Usage of **SVD 2nd time**, decomposing $\hat{F}$:
 - $F = U D^a V^T = U(\text{diag}(D_{11}, D_{22}, 0)) V^T$
 - with $\text{svd}(\hat{F}) = U D V^T$
 - and $D_{11} \geq D_{22} \geq D_{33}$

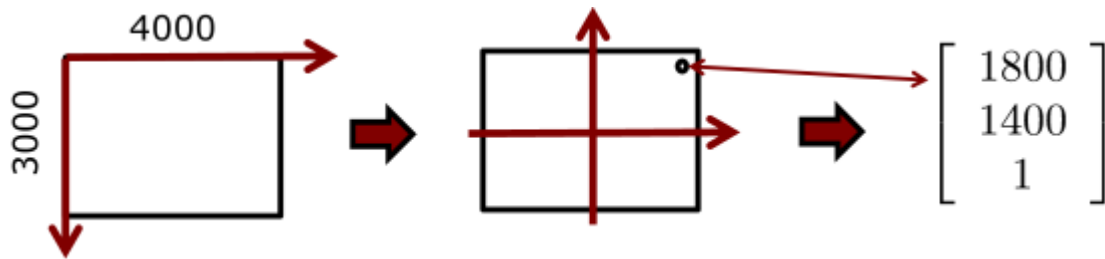
```

1 function F = F.from_point_pairs(xs, xss)
2 % xs, xss: Nx3 homologous point coordinates, N > 7
3 % F:      3x3 fundamental matrix
4
5 % coefficient matrix
6 for n = 1 : size(xs, 1)
7     A(n, :) = kron(xss(n, :), xs(n, :));
8 end
9
10 % singular value decomposition
11 [U, D, V] = svd(A);
12
13 % approximate F, possibly regular
14 Fa = reshape(V(:, 9), 3, 3);
15
16 % svd decomposition of F
17 [Ua, Da, Va] = svd(Fa);
18
19 % algebraically best F, singular
20 F = Ua * diag([Da(1, 1), Da(2, 2), 0]) * Va';

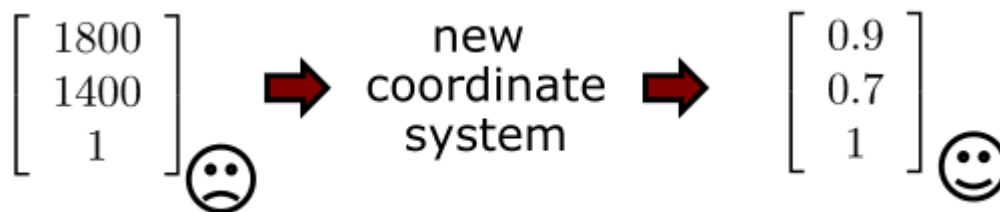
```

Conditioning/Normalization

- Well-Conditioned Problem
 - Example image 12MPixel camera



- III-conditioned, numerically stable
 - small distortions can have a huge effect



o

Normalisation: of point coordinate

- substantially **improves** stability
- Transformation:** center of mass of all points is at $(0, 0)$
- Scale** image: $x, y \in [-1, 1]$
- Define it as $T : \mathbf{T}\mathbf{x} = \hat{\mathbf{x}}$
- Determine fundamental matrix $\hat{F}$

$$\begin{aligned} \mathbf{x}'^T \mathbf{F} \mathbf{x}'' &= (T^{-1} \hat{\mathbf{x}}')^T F (T^{-1} \hat{\mathbf{x}}'') \\ &= \hat{\mathbf{x}}'^T T^{-1} F T^{-1} \hat{\mathbf{x}}'' \\ &= \hat{\mathbf{x}}'^T \hat{F} \hat{\mathbf{x}}'' \end{aligned}$$

- Obtain essential matrix F :

$$\begin{aligned} \hat{F} &= T^{-T} F T^{-1} \\ F &= T^T \hat{F} T \end{aligned}$$

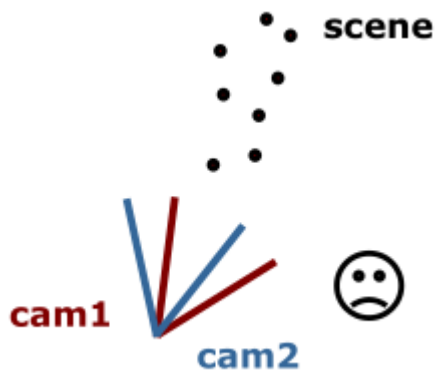
Singularity

- All corresponding points lie on plane: $\text{rank}(A) < 8$

- Numerically instable if points are close to a plane



- Projection centers of both cameras are identical:
- translation of the camera is zero between both images, i.e. : $\mathbf{X}_{O'} = \mathbf{X}_{O''}$



Essential Matrix

- Coplanarity constraint:

$$\circ \begin{bmatrix} {}^k x'_n & {}^k y'_n & c' \end{bmatrix} \underbrace{\begin{bmatrix} E_{11} & E_{12} & E_{13} \\ E_{21} & E_{22} & E_{23} \\ E_{31} & E_{32} & E_{33} \end{bmatrix}}_{\text{unknowns!}} \begin{bmatrix} {}^k x''_n \\ {}^k y''_n \\ c'' \end{bmatrix} = 0$$

```

1 function E = E_from_point_pairs(xs, xss)
2 % xs, xss: Nx3 homologous point coordinates, N > 7
3 % E:      3x3 essential matrix
4
5 % coefficient matrix
6 for n = 1 : size(xs, 1)
7     A(n, :) = kron(xss(n, :), xs(n, :));
8 end
9
10 % singular value decomposition
11 [U, D, V] = svd(A);
12
13 % approximate E, possibly regular
14 Ea = reshape(V(:, 9), 3, 3);
15
16 % svd decomposition of Ea
17 [Ua, Da, Va] = svd(Ea);
18
19 % algebraically best E, singular, same rank as Ea
20 E = Ua * diag([1, 1, 0]) * Va';

```

build matrix A

solve Ae=0

build matrix Ea

compute SVD of Ea

**build matrix E from Ea
by imposing constraints**

- Constraints: For essential matrix: both non-zero singular values are identical

$$E = U \begin{bmatrix} d & 0 & 0 \\ 0 & d & 0 \\ 0 & 0 & 0 \end{bmatrix} V^T = U \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} V^T$$

↑
homogenous

Conditioning and Normalisation

- Is essential: Transforming and scaling
 - Same approach as for F
- Determine fundamental matrix $\hat{E}$

$$\begin{aligned}
 {}^k\mathbf{x}'^T \mathbf{E} {}^k\mathbf{x}'' &= (T^{-1k} \hat{\mathbf{x}}')^T E (T^{-1k} \hat{\mathbf{x}}'') \\
 &= {}^k\hat{\mathbf{x}}'^T T^{-1} E T^{-1k} \hat{\mathbf{x}}'' \\
 &= {}^k\hat{\mathbf{x}}'^T \hat{E} {}^k\hat{\mathbf{x}}''
 \end{aligned}$$

- Obtain essential matrix E:

$$\begin{aligned}
 \hat{E} &= T^{-T} E T^{-1} \\
 F &= T^T \hat{E} T
 \end{aligned}$$

- Properties of Essential Matrix:
 - Homogeneous
 - Singular: $|E| = 0$ (determinant is zero)
 - Two identical non-zero singular values

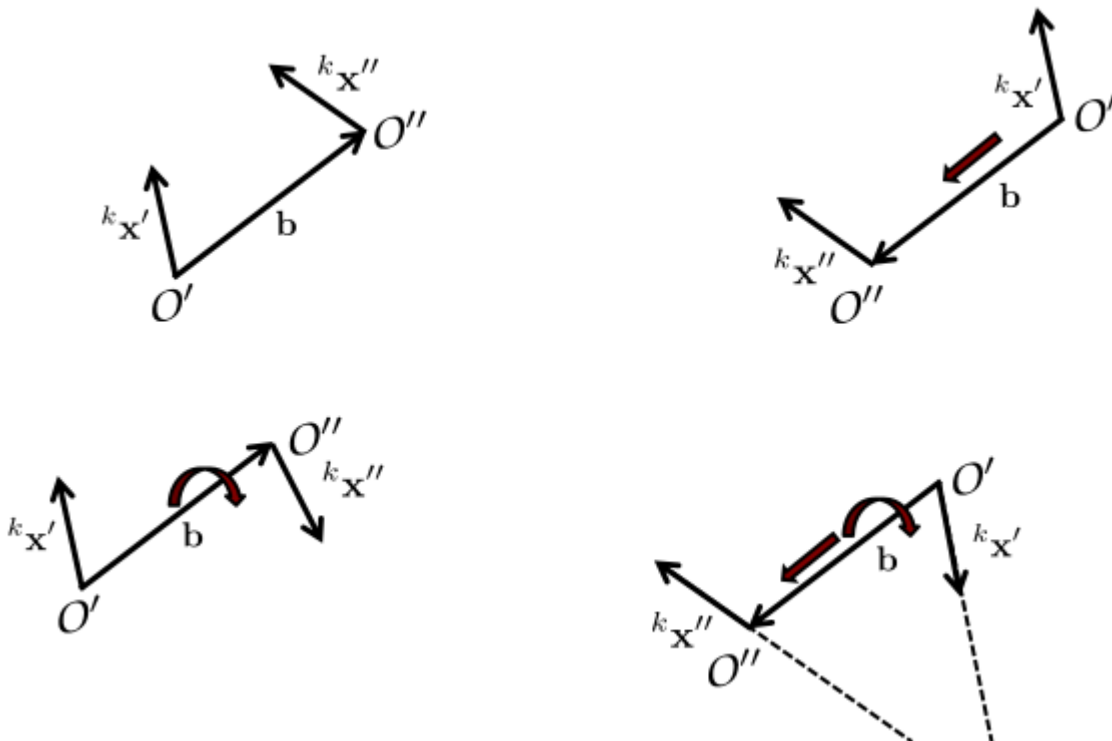
$$\circ E = U \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} V^T$$

5-Point Algorithm (Nistér 03/04)

- **Standard solution** for Direct solution
- Solving polynomial of degree 10
 - 10 possible solutions
- Often used together with RANSAC
 - Ransac proposes correspondences
 - Evaluate 5 point solutions based on other corresponding points
- Optional: Subsequent refinement using least squares only based on inlier points

Computing Orientation Parameters from Essential Matrix

- $E \rightarrow S_B, R$
- Multiple Solutions from Math
 - Multiplication of b up to a scalar
 - Rotation
 - Both



-
- Only physically plausible: **Points in front** of both cameras

Algebraic Solution (Hartley & Zissermann)

- For obtaining Basis and Rotation Matrix - Given Essential Matrix

$$\circ E = U \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} V^T$$

- with U, V as rotation matrices

- Define

$$\circ \text{Skew-Sym. Matrix: } Z = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\circ \text{Rotation matrix: } W = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\circ \text{So that } ZW = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

- Then

$$\begin{aligned} E &= UZ & WV^T \\ &= U \underbrace{ZU^T U}_I & WV^T \\ &= \underbrace{UZU^T}_{S_B} \underbrace{UWV^T}_{R^T} \end{aligned}$$

4 Possibilities to Define Z,W

$$\begin{aligned} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} &= ZW = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= -Z^T W = - \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}^T \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= -ZW^T = - \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}^T \\ &= Z^T W^T = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}^T \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}^T \end{aligned}$$

$$E = \underbrace{UZU^T}_{S_B} \underbrace{UWV^T}_{R^T}$$



2 solutions for S_B 2 solutions for R

$$S_{\hat{B}}^1 = UZU^T \quad S_{\hat{B}}^2 = UZ^T U^T \quad R_1^T = UWV^T \quad R_2^T = UW^T V^T$$


 4 solutions

$$E^1 = UZU^T UWV^T$$

$$E^2 = -UZ^T U^T UWV^T$$

$$E^3 = -UZU^T UW^T V^T$$

$$E^4 = UZ^T U^T UW^T V^T$$

Solution

- Compute the SVD of E: $UDV^T = \text{svd}(E)$
 - Normalise U, V by $U = U|U|, V = V|V|$
 - Compute 4 solutions:
 - $S_{\hat{B}}^1 = UZU^T$
 - $S_{\hat{B}}^2 = UZ^T U^T$
 - $R_1^T = UWV^T$
 - $R_2^T = UW^T V^T$
 - Test for which solution all points are in front of both cameras
 - Return physically plausible configuration
-
- Compute relative orientation from corresponding image points alone
 - Direct solution for F and E
 - Estimate camera motion (except scale)
 - Direct solutions
 - F/E from $N > 7$ points (8 Point Algorithm)
 - E from $N = 5$ points (Nistors 5-Point Algorithm)